Modern Quantum Mechanics J.J. Sakurai Solution Manual

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Chapter 1

Quantum Dynamics

1.1 Problem 2.1

$$H = -\frac{eB}{mc}S_z$$

$$\begin{aligned}
\frac{dA}{dt} &= \frac{1}{i\hbar}[A, H] + \frac{\partial A}{\partial t} \\
\frac{dS_x}{dt} &= \frac{1}{i\hbar}[S_x, \omega S_z] \\
&= \frac{\omega}{i\hbar}[S_x, S_z] \\
&= \frac{\omega}{i\hbar}i\hbar(-S_y)
\end{aligned}$$

$$\dot{S}_x = \omega S_y \tag{1.1}$$

Similarly,

$$\frac{dS_y}{dt} = \frac{1}{i\hbar}\omega[S_y, S_z]$$

$$\dot{S}_y = \omega S_x \tag{1.2}$$

Now, $\ddot{S}_x = -\omega\dot{S}_y = -\omega(\omega S_x)$

$$\ddot{S}_x + \omega^2 S_x = 0 \tag{1.3}$$

$$S_x(t) = A\cos\omega t + B\sin\omega t \tag{1.4}$$

$\ddot{S}_y = \omega\dot{S}_x = \omega(-\omega S_y)$

$$\ddot{S}_y + \omega^2 S_y = 0 \tag{1.5}$$

$$S_y(t) = C\cos\omega t + D\sin\omega t \tag{1.6}$$

using initial conditions at $t = 0$

$$S_x(0) = S_x \qquad S_y(0) = S_y$$

we can write,

$$S_x(t) = S_x cos\omega t + B \sin \omega t \tag{1.7}$$

$$S_y(t) = S_y cos\omega t + D \sin \omega t \tag{1.8}$$

But from (1.1)

$$-\omega S_x \sin \omega t + \omega B \cos \omega t = -\omega S_y \cos \omega t - \omega D \sin \omega t$$

$$B = -S_y \qquad D = S_x$$

$$\boxed{S_x(t) = S_x cos\omega t - S_y \sin \omega t} \tag{1.9}$$

$$\boxed{S_y(t) = S_y cos\omega t + S_x \sin \omega t} \tag{1.10}$$

$$\frac{dS_z}{dt} = \frac{\omega}{i\hbar}[S_z, S_z] = 0$$

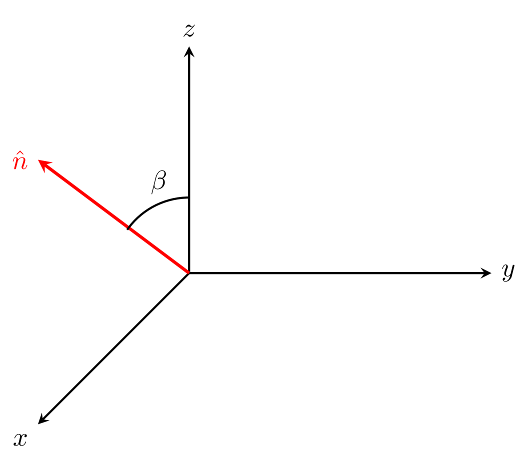
$$S_z(t) = C$$

at t=0 $S_z(0) = S_z$

$$\boxed{S_z(t) = S_z} \tag{1.11}$$

1.2 Problem 2.3

(a)



Choose a unit vector

$$\hat{n} = \sin\beta\hat{x} + \cos\beta\hat{z} \tag{1.12}$$

So that,

$$\vec{S}.\hat{n} = \frac{\hbar}{2}\sin\beta\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} + \frac{\hbar}{2}\cos\beta\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$$

$$\vec{S}.\hat{n} = \frac{\hbar}{2}\begin{pmatrix}\cos\beta & \sin\beta\\ \sin\beta & -\cos\beta\end{pmatrix} \tag{1.13}$$

The eigenvalues are $\pm\frac{\hbar}{2}$
Eigenvector corresponding to $\frac{\hbar}{2}$ is $\begin{pmatrix}\cos\frac{\beta}{2}\\ \sin\frac{\beta}{2}\end{pmatrix}$ therefore,

$$\left|\vec{S}.\hat{n};+\right\rangle = \cos\frac{\beta}{2}\left|+\right\rangle \sin\frac{\beta}{2}\left|-\right\rangle \tag{1.14}$$

Now, $\hat{H} = \omega S_z$

$$\begin{aligned}\left|\vec{S}.\hat{n};+;t\right\rangle &= \exp\left(-\frac{iHt}{\hbar}\right)\left|\vec{S}.\hat{n};+\right\rangle\\ &= e^{-\frac{i\omega t}{2}}\cos\frac{\beta}{2}\left|+\right\rangle + e^{\frac{i\omega t}{2}}\sin\frac{\beta}{2}\left|-\right\rangle\end{aligned} \tag{1.15}$$

Now probability of finding $s_x = \hbar/2$ will be

$$\left|\left\langle S_x,+\middle|\vec{S}.\hat{n},+;t\right\rangle\right|^2$$

$$\begin{aligned}\left\langle S_x,+\middle|\vec{S}.\hat{n},+;t\right\rangle &= \frac{1}{\sqrt{2}}\left(\left\langle +\right| + \left\langle -\right|\right)\left(e^{-i\omega t/2}\cos\frac{\beta}{2}\left|+\right\rangle + e^{i\omega t/2}\sin\frac{\beta}{2}\left|-\right\rangle\right)\\ &= \frac{1}{\sqrt{2}}\left(e^{-i\omega t/2}\cos\frac{\beta}{2} + e^{i\omega t/2}\sin\frac{\beta}{2}\right)\end{aligned}$$

$$\begin{aligned}\left|\left\langle S_x,+\middle|\vec{S}.\hat{n},+;t\right\rangle\right|^2 &= \frac{1}{2}\left(e^{-i\omega t/2}\cos\frac{\beta}{2} + e^{i\omega t/2}\sin\frac{\beta}{2}\right)\times\\ &\qquad \left(e^{i\omega t/2}\cos\frac{\beta}{2} + e^{-i\omega t/2}\sin\frac{\beta}{2}\right)\\ &= \frac{1}{2}\left[1 + \sin\frac{\beta}{2}\cos\frac{\beta}{2}\left(e^{i\omega t/2} + e^{-i\omega t/2}\right)\right]\end{aligned}$$

$$\boxed{\left|\left\langle S_x,+\middle|\vec{S}.\hat{n},+;t\right\rangle\right|^2 = \frac{1}{2}\left(1 + \sin\beta\cos\omega t\right)} \tag{1.16}$$

(b)

$$\left|\vec{S}.\hat{n},+;t\right\rangle = \begin{pmatrix} e^{-i\omega t/2}\cos\frac{\beta}{2} \\ e^{i\omega t/2}\sin\frac{\beta}{2} \end{pmatrix}$$

let $|\alpha\rangle = \left|\vec{S}.\hat{n},+;t\right\rangle$
$\langle S_x\rangle = \langle\alpha| S_x |\alpha\rangle$

$$\begin{aligned} S_x|\alpha\rangle &= \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^{-i\omega t/2}\cos\frac{\beta}{2} \\ e^{i\omega t/2}\sin\frac{\beta}{2} \end{pmatrix} \\ &= \frac{\hbar}{2}\begin{pmatrix} e^{i\omega t/2}\sin\frac{\beta}{2} \\ e^{-i\omega t/2}\cos\frac{\beta}{2} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \langle S_x\rangle &= \frac{\hbar}{2}\begin{pmatrix} e^{i\omega t/2}\cos\frac{\beta}{2} & e^{-i\omega t/2}\sin\frac{\beta}{2} \end{pmatrix}\begin{pmatrix} e^{i\omega t/2}\sin\frac{\beta}{2} \\ e^{-i\omega t/2}\cos\frac{\beta}{2} \end{pmatrix} \\ &= \frac{\hbar}{2}\cos\frac{\beta}{2}\sin\frac{\beta}{2}\left(e^{i\omega t}+e^{-i\omega t}\right) \end{aligned}$$

$$\boxed{\langle S_x\rangle = \frac{\hbar}{2}\sin\beta\cos\omega t} \tag{1.17}$$

(c)

(i)

As $\beta \to 0$ $\quad \langle S_x\rangle \to 0$

(ii)

As $\beta \to \frac{\pi}{2}$ $\quad \langle S_x\rangle \to \cos\omega t$

As Expected

1.3 Problem 2.4

$$|\nu_e\rangle = \cos\theta|\nu_1\rangle - \sin\theta|\nu_2\rangle \tag{1.18}$$

$$U(t,0)|\nu_e\rangle = |\nu_e,t\rangle$$
$$|\nu_e,t\rangle = e^{-iE_1t/\hbar}\cos\theta|\nu_1\rangle - e^{-iE_2t/\hbar}\sin\theta|\nu_2\rangle$$
$$\langle\nu_e|\nu_e;t\rangle = e^{-iE_1t/\hbar}\cos^2\theta - e^{-iE_2t/\hbar}\sin^2\theta$$

$$\begin{aligned} |\langle\nu_e|\nu_e;t\rangle|^2 = &\left(e^{iE_1t/\hbar}\cos\theta\langle\nu_1| - e^{iE_2t/\hbar}\sin\theta\langle\nu_2|\right)\times \\ &\left(e^{-iE_1t/\hbar}\cos\theta|\nu_1\rangle - e^{-iE_2t/\hbar}\sin\theta|\nu_2\rangle\right) \end{aligned}$$

1.4 Problem 2.5

$$\begin{aligned}\frac{d}{dt}x(t) &= \frac{1}{i\hbar}\left[x, \frac{p^2}{2m}\right] \\ &= \frac{1}{2m}[x,p] = \frac{p}{m}\end{aligned}$$

therefore,

$$x(t) = x(0) + \frac{pt}{m} \tag{1.19}$$

$$\begin{aligned}[x(t), x(0)] &= [x(0), x(0)] + \left[\frac{pt}{m}, x(0)\right] \\ &= -\frac{t}{m}[x,p]\end{aligned}$$

$$\boxed{[x(t), x(0)] = -\frac{i\hbar t}{m}} \tag{1.20}$$

1.5 Problem 2.6

$$\begin{aligned}[[H,x],x] &= \left[\left[\frac{p^2}{2m} + V(x), x\right], x\right] \\ &= \frac{1}{2m}\left[\left[p^2, x\right], x\right] \\ &= \frac{1}{2m}\left[-i\hbar 2p, x\right] \\ &= \frac{i\hbar}{m}\left[x, p\right]\end{aligned}$$

$$[[H,x],x] = -\frac{\hbar^2}{m} \tag{1.21}$$

$$\begin{aligned}
&\sum_{a'} |\langle a''| x |a'\rangle|^2 (E_{a'} - E_{a''}) \\
&= \sum_{a'} \langle a''| x |a'\rangle \langle a'| x |a''\rangle (E_{a'} - E_{a''}) \\
&= \sum_{a'} E_{a'} \langle a''| x |a'\rangle \langle a'| x |a''\rangle - \sum_{a'} E_{a''} \langle a''| x |a'\rangle \langle a'| x |a''\rangle \\
&= \sum_{a'} \langle a''| xH |a'\rangle \langle a'| x |a''\rangle - E_{a''} \langle a''| x^2 |a''\rangle \\
&= \langle a''| xHx |a''\rangle - \frac{1}{2} \left(\langle a''| x^2 H |a''\rangle + \langle a''| Hx^2 |a''\rangle \right) \\
&= -\frac{1}{2} \left(\langle a''| x^2 H + Hx^2 - 2xHx |a''\rangle \right) \\
&= -\frac{1}{2} \langle a''| [[H, x], x] |a''\rangle \\
&= -\frac{1}{2} \left(-\frac{\hbar^2}{m} \right)
\end{aligned}$$

$$\boxed{\sum_{a'} |\langle a''| x |a'\rangle|^2 (E_{a'} - E_{a''}) = \frac{\hbar^2}{2m}} \tag{1.22}$$

1.6 Problem 2.7

$$\begin{aligned}
[\vec{x}.\vec{p}, H] &= \vec{x}.[\vec{p}, H] + [\vec{x}, H].\vec{p} \\
&= \vec{x}.\left[\vec{p}, \frac{p^2}{2m} + V(x)\right] + \left[\vec{x}, \frac{p^2}{2m} + V(x)\right].\vec{p} \\
&= \vec{x}.[\vec{p}, V(x)] + \frac{1}{2m}\left[\vec{x}, p^2\right].\vec{p} \\
&= i\hbar\vec{x}.(-\nabla V) + \frac{2i\hbar}{2m}\vec{p}.\vec{p}
\end{aligned}$$

$$[\vec{x}.\vec{p}, H] = i\hbar\left(\frac{p^2}{2m} - \vec{x}.\nabla V\right) \tag{1.23}$$

$$\begin{aligned}
\frac{d}{dt}\langle\vec{x}.\vec{p}\rangle &= \frac{1}{i\hbar}\langle[\vec{x}.\vec{p}, H]\rangle + \frac{\partial}{\partial t}\langle\vec{x}.\vec{p}\rangle \\
&= \left\langle \frac{p^2}{2m} - \vec{x}.\nabla V \right\rangle
\end{aligned}$$

$$\boxed{\frac{d}{dt}\langle\vec{x}.\vec{p}\rangle = \left\langle\frac{p^2}{2m}\right\rangle - \left\langle\vec{x}.\nabla V\right\rangle} \tag{1.24}$$

1.7 Problem 2.18

Ground state of 1-D Simple Harmonic Oscillator is given by

$$\psi_0(x,t) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{m\omega x^2}{2\hbar} - i\frac{E_0 t}{\hbar}\right) \tag{1.25}$$

with $E_0 = \frac{\hbar\omega}{2}$. Now the correlation function

$$\begin{aligned}
C(t) &= \langle x(t)x(0)\rangle \\
&= \langle 0|\left(x(0)\cos\omega t + \left[\frac{p(0)}{m\omega}\right]\sin\omega t\right) x\,|0\rangle \\
&= \langle 0|\sqrt{\frac{\hbar}{2m\omega}}\left(x(0)\cos\omega t + \left[\frac{p(0)}{m\omega}\right]\sin\omega t\right)|1\rangle \\
&= \langle 0|\sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{\frac{\hbar}{2m\omega}}\cos\omega t\left(\sqrt{2}\,|2\rangle + |0\rangle\right) + \frac{i}{m\omega}\sqrt{\frac{m\hbar\omega}{2}}\sin\omega t\left(\sqrt{2}\,|2\rangle - |0\rangle\right)\right) \\
&= \frac{\hbar}{2m\omega}\left(\cos\omega t - i\sin\omega t\right)
\end{aligned}$$

$$\boxed{C(t) = \frac{\hbar}{2m\omega}e^{-i\omega t}} \tag{1.26}$$

1.8 Problem 2.19

(a)

let us construct a state

$$|\psi\rangle = c\,|0\rangle + d\,|1\rangle \tag{1.27}$$

where c and d are complex. Now,

$$\begin{aligned}
\langle x\rangle &= \sqrt{\frac{\hbar}{2m\omega}}\left(c^*\langle 0| + d^*\langle 1|\right)\left(a + a^\dagger\right)\left(c\,|0\rangle + d\,|1\rangle\right) \\
&= \sqrt{\frac{\hbar}{2m\omega}}\left(c^*\langle 0| + d^*\langle 1|\right)\left(d\,|0\rangle + c\,|1\rangle + \sqrt{2}\,|2\rangle\right) \\
&= \sqrt{\frac{\hbar}{2m\omega}}\left(c^*d + cd^*\right)
\end{aligned}$$

for this to be real c and d should be purely real or imaginary. We take them to be real[1]. Also we have constrain

$$c^2 + d^2 = 1$$

[1]Even if we take them imaginary it would only add a phase factor

And therefore,

$$\begin{aligned}
\langle x\rangle &= \sqrt{\frac{\hbar}{2m\omega}}2cd \\
&= \sqrt{\frac{2\hbar}{m\omega}}c\sqrt{1-c^2}
\end{aligned}$$

Now finding maximum of this function we get $c = \pm 1/\sqrt{2}$ and hence $d = \pm 1/\sqrt{2}$. So we get a state

$$\boxed{|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)} \tag{1.28}$$

(b)

$$|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}}|\psi(0)\rangle$$

$$\boxed{|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-\frac{i\omega t}{2}}|0\rangle + e^{-\frac{i3\omega t}{2}}|1\rangle\right)} \tag{1.29}$$

(i)

$$\begin{aligned}
\langle x\rangle &= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(e^{\frac{i\omega t}{2}}\langle 0| + e^{-\frac{i3\omega t}{2}}\langle 1|\right)\left(a + a^\dagger\right)\times \\
&\quad \left(e^{-\frac{i\omega t}{2}}|0\rangle + e^{-\frac{i3\omega t}{2}}|1\rangle\right) \\
&= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(e^{\frac{i\omega t}{2}}\langle 0| + e^{\frac{i3\omega t}{2}}\langle 1|\right)\times \\
&\quad \left(e^{-\frac{i\omega t}{2}}|1\rangle + e^{-\frac{i3\omega t}{2}}\sqrt{2}\,|2\rangle + e^{-\frac{i3\omega t}{2}}|0\rangle\right) \\
&= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(e^{-i\omega t} + e^{i\omega t}\right)
\end{aligned}$$

$$\boxed{\langle x\rangle = \sqrt{\frac{\hbar}{2m\omega}}\cos\omega t} \tag{1.30}$$

(ii)

$$\frac{d\langle x\rangle}{dt} = \frac{1}{i\hbar}\langle [x, H]\rangle + \frac{\partial\langle x\rangle}{\partial t}$$

Chapter 2

Approximation Methods

2.1 Problem 5.30

(a)

We have $F(t) = F_0 e^{-t/\tau}$ hence the potential,

$$V = -\int F dx = -F_0 e^{-t/\tau} x \tag{2.1}$$

- $c_1^{(0)}(t) = \delta_{10} = 0$

- $$\begin{aligned}
c_1^{(1)}(t) &= \frac{-i}{\hbar} \int_0^t dt' e^{i\omega_{10}t'} V_{10}(t') \\
&= \frac{-i}{\hbar} \int_0^t dt' e^{i\omega t'} \langle 1| \left(-F_0 e^{-t/\tau}\right) \sqrt{\frac{\hbar}{2m\omega}} \left(a + a^\dagger\right) |0\rangle \\
&= \frac{i}{\hbar} \sqrt{\frac{\hbar}{2m\omega}} \int_0^t e^{(i\omega - 1/\tau)t'} dt' \\
&= \frac{i}{\hbar} \sqrt{\frac{\hbar}{2m\omega}} \frac{e^{(i\omega-1/\tau)t} - 1}{i\omega - \frac{1}{\tau}} \\
&= \frac{-\tau F_0}{\sqrt{2m\hbar\omega}} \frac{1 - e^{(i\omega-1/\tau)t}}{\tau\omega + i}
\end{aligned}$$

$$
\begin{aligned}
|c_1^{(1)}(t)|^2 &= \frac{\tau^2 F_0^2}{2m\hbar\omega}\frac{1}{1+(\tau\omega)^2}\left(1-e^{(-i\omega-1/\tau)t}\right)\left(1-e^{(i\omega-1/\tau)t}\right) \\
&= \frac{\tau^2 F_0^2}{2m\hbar\omega}\frac{1}{1+(\tau\omega)^2}\left[1+e^{-2t/\tau}-e^{-t/\tau}\left(e^{i\omega t}+e^{-i\omega t}\right)\right] \\
&= \frac{\tau^2 F_0^2}{2m\hbar\omega}\frac{1}{1+(\tau\omega)^2}\left[1+e^{-2t/\tau}+e^{-t/\tau}-2e^{-t/\tau}\cos\omega t\right]
\end{aligned}
$$

Therefore probability of finding oscillator in first excited state is

$$
\boxed{P(0\to 1) = \frac{\tau^2 F_0^2}{2m\hbar\omega}\frac{1}{1+(\tau\omega)^2}\left[1+e^{-2t/\tau}+e^{-t/\tau}-2e^{-t/\tau}\cos\omega t\right]} \tag{2.2}
$$

As $t\to\infty$

$$
\boxed{P(0\to 1) = \frac{\tau^2 F_0^2}{2m\hbar\omega}\frac{1}{1+(\tau\omega)^2}} \tag{2.3}
$$

This is surprising as after long time the force $F(t\to\infty)=0$ and hence system should behave as if it was unperturbed and should be in ground state which contradict with non-zero probability of finding system in first excited state.

(b)

- For higher excites states in coefficient $c_n^{(1)}(t)$ we'll have term

  $$\langle n|\left(a+a^\dagger\right)|0\rangle$$

  Which will be 0 for $n\geq 2$. Hence, upto first order corrections we cannot find higher excited states.

- If we go to second order correction we would have term

  $$\sum_m \langle n|\left(a+a^\dagger\right)|m\rangle\langle m|\left(a+a^\dagger\right)|0\rangle$$

  which allows us to have only $m=1$ and hence $n=2$ for non-vanishing term.

$$\boxed{\text{We can find higher excited states but they are less likely}}$$